

Report on the outcomes of a Short-Term Scientific Mission¹

Action number: CA21129

Grantee name: Jana Bernhard-Harrer

Details of the STSM

Title: AmCAT for MEDem: Developing Open Science Infrastructure for Opinionated Text Analysis (STSM)

Description of the work carried out during the STSM

The Short-Term Scientific Mission (STSM) was conducted at Vrije Universiteit Amsterdam in collaboration with the Text and Multimedia Data Service Centre. The mission focused on integrating and deploying the AmCAT infrastructure within the MEDem (Monitoring Electoral Democracy) environment, while simultaneously advancing open science standards and methodological development within the COST Action OPINION framework.

During the first phase, technical setup and testing were prioritised. A dedicated test server was configured to install and systematically evaluate AmCAT under different conditions. This included testing installation procedures, deletion, reinstallation, and software updates to ensure stability and reproducibility. Based on these experiments, detailed step-by-step documentation was written, specifically tailored for users with limited experience in server administration and command-line environments. These notes were immediately published as a blog entry (https://societal-analytics.nl/blogs/20260219_intall_amcat_server/), partly integrated into AmCAT's GitHub README files and will serve as the basis for a forthcoming contribution to the AmCAT Book (<https://amcat.nl/book/>).

Following the successful test deployment, an operational AmCAT instance was installed on a Vienna-based server maintained by the Computational Communication Science Lab (CCL) for the MEDem project. The installation was carried out in close coordination with the Viennese IT department to ensure compliance with institutional security and maintenance standards. The instance is fully operational and prepared for branding and user access configuration.

In parallel to the technical work, conceptual and methodological discussions were held regarding interoperability between AmCAT and the MEDem "Text as Data" model, particularly with regard to

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compatibility with the MEDem ATLAS prototype. These discussions identified concrete feature extensions that would enhance cross-platform interoperability and support comparative multilingual research.

The second phase of the STSM focused on dissemination and methodological reflection. An introduction and structured outline for a collaborative academic article on open science content analysis with AmCAT were drafted and circulated among co-authors for further development. In addition, preliminary text was prepared for a the amcat book chapter describing best practices for setting up and maintaining institutional AmCAT servers.

The visit also included a presentation at the Political Communication Lab meeting, where ongoing research on sentiment toward political actors in news media was discussed. A final meeting with the host team focused specifically on usability issues from a non-technical researcher's perspective, generating practical recommendations for improving accessibility.

No major deviations from the original work plan occurred. However, greater emphasis was placed on user experience evaluation and interoperability discussions than initially anticipated, reflecting the practical insights gained during implementation.

Description of the STSM main achievements and planned follow-up activities

The STSM successfully achieved its planned goals and produced tangible technical, methodological, and collaborative outcomes that directly contribute to the objectives of COST Action OPINION.

First, the infrastructure development objective was fully realised. A stable and reproducible AmCAT installation was successfully deployed on MEDem servers in Vienna. The instance is operational and ready for branding and user onboarding. In addition, systematic testing on a separate server ensured that installation, deletion, and updating procedures are now clearly documented and replicable. This significantly lowers technical barriers for future institutional deployments and supports OPINION's aim of establishing shared computational foundations for opinion analysis across countries and languages.

Second, the STSM substantially contributed to capacity building and open science integration. Comprehensive technical documentation was produced, including beginner-friendly installation instructions and maintenance guidelines. These materials have already been incorporated into the AmCAT GitHub repository and will form the basis of a detailed handbook chapter. By standardising setup procedures and emphasising transparency, version control, and reproducibility, the mission strengthens WG3's work on benchmarked, interoperable, and open research workflows.

Third, the mission advanced knowledge exchange and interdisciplinary collaboration. Close cooperation with colleagues at Vrije Universiteit Amsterdam fostered alignment between AmCAT development and MEDem's evolving "Text as Data" model. Discussions on interoperability with the MEDem ATLAS prototype identified concrete feature developments that will enhance cross-project integration. These exchanges directly support OPINION's objective of reducing fragmentation across research communities and establishing cumulative methodological standards.

Fourth, dissemination goals were initiated successfully. A structured introduction and outline for a collaborative journal article on "Open Science Content Analysis with AmCAT" were drafted and circulated among co-authors. This article will present AmCAT as an accessible infrastructure for transparent and reproducible computational content analysis. In parallel, work began on a chapter for the AmCAT book contribution detailing institutional server setup and maintenance procedures. Both outputs will provide widely accessible methodological resources for the research community (see first blog post here: https://societal-analytics.nl/blogs/20260219_intall_amcat_server/).

Overall, the STSM made a measurable contribution to OPINION's core objectives: strengthening conceptual and technical foundations, enabling multilingual comparative research, enhancing research capacity, and embedding open science practices within computational communication research.

Future collaboration has already been agreed upon. The handbook chapter and journal article will be finalised jointly with the host team. Continued exchange will focus on feature development for interoperability with MEDem ATLAS. The installation guide and documentation will be accessible for anyone interested in learning about AmCAT.

In sum, the STSM not only achieved its planned outcomes but also laid the groundwork for sustained infrastructure development and collaborative research within OPINION.